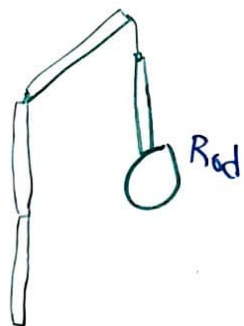
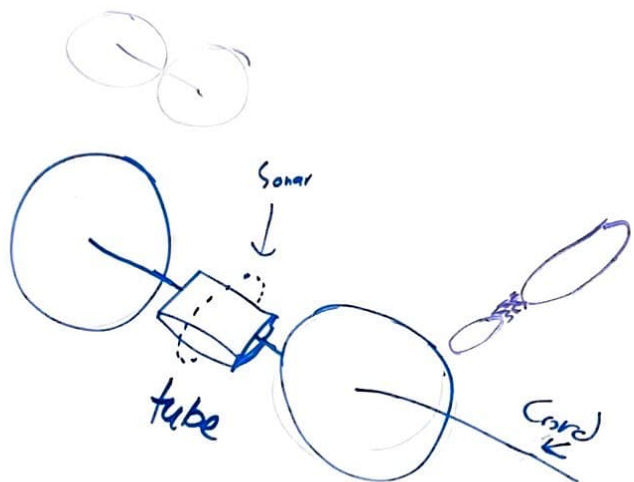


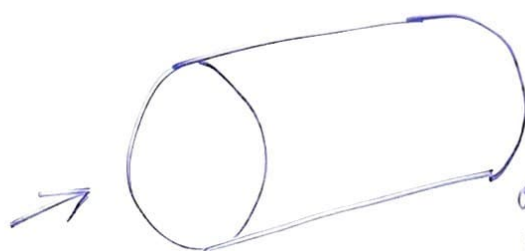
Natural Gas

- More difficult to replicate environment



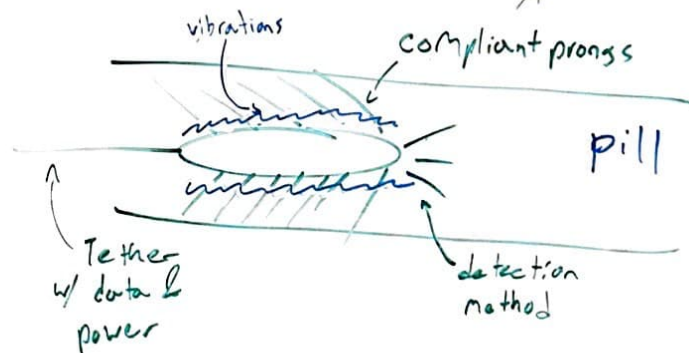
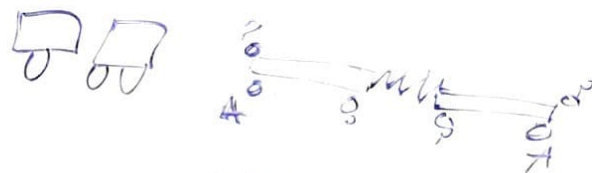
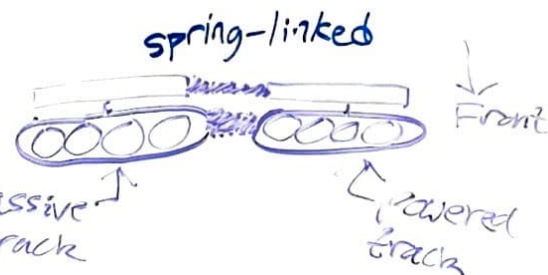
Conduit

- Typically too SM 422



Fluid

- Wireless Communication



Step 2:

Alessandro → - What is the goal?
Julian → - Who is the audience

Step 3:

Cues?

durability,
safety, comfort.

Device instruction

Issues to Detect

- Cracks
- Debris
- Clogs
- Penetration

Ideas for locating objects

- Encoders on wheels
- Camera / Sonar
- Tether to base
- Wired preferred

Goals

- Detection
- Retrievability
- Connection

Traveling Methods

- DC Wheels
- Mecanum Wheels
- Vibration
- Tank tread

Fault Detection

- Camera Vision
- Lidar
- Ultrasonic
- Physical
- Infrared

Rank/sort

Traversal:

- 1) Push rod
- 2) Powered wheels
- 3) Vibration
- 4) Tank tread
- 5) Magnetic
- 6) Pull

Fault Detection:

- 1) Camera w. flash
- 2) IR
- 3) Sonar
- 4) "Whisker"
force sensor
- 5) LIDAR

Location Detection

- 1) Wheel encoders
- 2) Tethered rope markers
- 3) Transmitter pylons
- 4) Signal bouncing
- 5) GPS

Body-Type

- 1) Spring-linked
- 2) Pill
- 3) Tube
- 4) Rod
- 5) Ball